IMPULSE V1: Quantifying Systematic Limits of TLE-Based Passive Orbital Sensing for Microdebris Environment Model Discrimination

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Abstract

The sub-catalog (10 cm or less) orbital debris environment is poorly constrained. NASA's ORDEM 3.0 and ESA's MASTER 8 debris-environment models diverge by more than two orders of magnitude in key LEO regimes (Manis et al., 2021). Despite this, independent observational constraints are scarce. IMPULSE (**I**nferred **M**icrodebris **P**opulation from **U**nmaneuvered **L**EO **S**tate **E**volution) uses decommissioned non-maneuvering LEO satellites as passive sensors, using historical Two-Line Element (TLE) data to detect statistically significant orbital deviations that could be candidates for microdebris induced impulses. The detection pipeline uses 9-state augmented Extended Kalman Filter with a Fraser-Potter fixed-interval smoother to isolate candidate perturbations. Non-singular equinoctial state variables and quality cuts were implemented to avoid numerical instability. The pipeline was validated through a four-stage synthetic ladder (null test, bias recovery, synthetic Δv injection, and clean-vs-debris detection), which provided important calibration, background subtraction, efficiency and error quantification numbers. IMPULSE achieves a 50% detection efficiency at $\Delta v_{50} = 0.041$ m/s on synthetic data and $\Delta v_{50} = 0.0385$ on real TLEs. Applied to the full fleet (3,325 decommissioned spacecraft and 15,830 valid analysis windows), the pipeline yields 8,161 candidate events (51.55% trigger rate). After background subtraction via a real TLE scrambled-residual null test (19.46% false positive rate), the net excess is $5,080 \pm 103$ events, which is 32.09% per window. This excess is significantly higher than the predictions made by ORDEM (847.9 detectable events or 5.36%) and MASTER (6.1 detectable events or 0.04%). The implied non-debris systematic floor is therefore 26-32%, preventing ORDEM/MASTER discrimination under the baseline operating regime. This result establishes the information limit of TLE-based passive sensing under the baseline model and charts a V2 path towards breaking that limit.

1 Introduction

1.1 Sub-Catalog Orbital Debris

Sub-catalog debris is the population of space debris that is too small to be tracked and logged in official space object directories. Typically, this line is drawn around 10 cm or smaller diameter objects (Liou, 2020; Manis et al., 2023). That is also the threshold that defines IMPULSE's regime of relevance: < 10 cm. Conventional space-surveillance approaches that use radar or optical tracking cannot maintain a continuous and reliable catalog of debris at this scale due to their sensitivity and noise floors (Manis et al., 2023). Yet, this population of debris poses a significant threat when considered in hypervelocity. While smaller, there is far more sub-catalog debris than the cataloged population. Depending on the size, cumulative tiny impacts in hypervelocity could result in change in orbital trajectories and cause surface degeneration of spacecraft (Sagnières & Sharf, 2017; Ward et al., 2023). In rarer cases, larger sized debris could potentially damage critical subsystems.

1.2 Existing Debris-Environment Models

Since the sub-catalog population cannot be directly observed and tracked, statistical characterization is necessary at this scale. The two major debris-environment models are NASA's Orbital Debris Engineering Model (ORDEM 3.0) and ESA's Meteoroid and Space Debris Terrestrial Environment Reference (MASTER 8) (Braun, Horstmann, et al., 2021; Krisko, 2014). These models use different input sources, observational data, assumptions and models. While they agree in some regimes, particularly in the cataloged population, they can diverge dramatically in the sub-catalog regime. This divergence can be up to over two orders of magnitude. In the regime that IMPULSE operates in, the divergence is $\sim 138\times$ (Manis et al., 2021). The implications of this go beyond the scientific communities, as these models quantify a debris population that is dangerous enough to be considered while designing spacecraft missions. The weighing of one model over the other could mean exaggerated mission costs and attention towards orbital debris protection, or the underprotection of vital subcomponents (Christiansen, 2009). As such, independent validations are necessary to tighten the uncertainty between the two models.

However, attempts to do so have been largely limited in their scope. Returned surfaces can provide ground-truth measurements, but they only represent specific epochs in time (Stansbery & Foster, 2005). Dedicated impact detectors like DEBIE produce very good time resolution, but are limited to their trajectories, cost and rarity (Kuitunen et al., 2001). Optical and radar observation approaches can detect small debris in aggregate, although they are dependent on the weather, geographic restrictions and have an ungenerous noise floor (Manis et al., 2023).

1.3 Passive Orbital Sensing and the IMPULSE Approach

A different, comparatively less explored option is to study the past orbital behaviour of spacecraft that already exist. Instead of directly observing debris, this approach looks for effects of debris-induced perturbations in the orbital histories of spacecraft. Modern tracking infrastructure uses Two-Line Element (TLE)

data to track and describe spacecraft orbiting around the Earth. Existing work in this approach has largely focused on approaching orbital anomalies as individual discrete events, and testing the detection of synthetically injected impulses within osculating elements (Bennett et al., 2023). IMPULSE builds its base off of studying spacecraft orbital history for deviations as a form of debris detection. However, IMPULSE modifies its goal to be used in real TLE histories, which operates in the mean-element space rather than the typically studied osculating element space (Hoots & Roehrich, 1980). Additionally, the detector looks for real orbital perturbations rather than synthetically injected impulses.

In order to isolate the candidate perturbations from burns and attitude correction maneuvers, IMPULSE specifically uses TLE histories of decommissioned satellites. Using a fleet of these satellites would provide significantly more statistical information than tracking one spacecraft. Aggregating the candidate perturbations from each spacecraft's TLE history and comparing it to the expected number of perturbation events under ORDEM or MASTER environments would then allow the pipeline to potentially discriminate between the two, provided enough statistical power. As such, IMPULSE is not a tool to reconstruct the debris population, but rather a discrimination study between two competing models, ORDEM and MASTER.

A closely related line of work is the synthetic-injection study of Bennett et al. (2023), which demonstrated that statistically significant orbital perturbations can be recovered from high-fidelity osculating-element trajectories under controlled conditions (Bennett et al., 2023). IMPULSE builds directly on that foundation but shifts the problem into the regime that actually exists in the historical record: real Two-Line Element (TLE) histories expressed in mean elements. This change introduces a substantially different measurement-error environment, catalog artifacts, and force-model mismatches that are absent from purely synthetic osculating-element experiments. The present work therefore constitutes a first attempt to test whether the passive-sensing concept remains viable once these real-world complications are included.

1.4 Research Objective

The objective of IMPULSE is to use decommissioned LEO spacecraft as passive dynamic sensors in order to characterize the sub-catalog regime enough to be able to discriminate between the ORDEM and MASTER models. IMPULSE uses a fleet of such spacecraft in order to maximize the statistical power. Each spacecraft is then run through a 9-state augmented Extended Kalman Filter forwards and backwards, which is smoothed through a Fraser-Potter fixed-interval smoother (Fraser & Potter, 1969). The smoothed output is examined for candidate perturbations.

A four-stage validation ladder is used to verify, quantify and calibrate the model and its results. Null tests quantify the false-positive rates of intrinsically, externally by confounders, and TLE-related errors. A bias test is used to study the model's response to long, slow-decaying forces and to tune model parameters to strike a balance between observing impulsive events and suppressive slow-decaying effects. The efficiency of the model is quantified via injection-and-recovery tests, giving detection efficiency curves for both MASTER and ORDEM environment, as well as calibrating the detection threshold for candidate perturbations. The model was also run on real-data and compared to synthetic runs to quantify the pipeline's unmodeled sources of error and the noise floor.

2 Research Question and Experimental Design

2.1 Research Questions and Measurement Hypothesis

The primary question IMPULSE aims to answer is: to what extent can the historical TLE orbital-state data from decommissioned, non-maneuvering LEO satellites be used as passive measurements of sub-catalog debris impacts? Additionally, do the measurements have enough statistical power to discriminate between ORDEM and MASTER's predictions? The question is built on the hypothesis that debris impacts impart enough impulse to produce small transient orbital-state perturbations that survive conversion to mean-element space when read as TLEs (Bennett et al., 2023). While the effects are tiny in the 1 mm to 10 cm range, a sufficiently precise force model may allow debris-induced perturbations to be distinguished statistically from the background of other unmodeled effects. Additionally, background-subtractions of false-positives will help isolate the debris impacts. The central difficulty lies in the unmodeled perturbations in the TLE histories that are not accounted for in the model or the background subtraction process. Outside of external forces like drag, solar radiation pressure, and geomagnetic storms, this primarily includes TLE-cataloging error, such as discontinuities and station changing (Osweiler, 2006; Vallado & Cefola, 2012). A successful measurement would distinguish a statistically significant debris-induced signal from external forces, modeling errors, and TLE-inherent cataloging errors.

2.2 Passive Dynamical Sensor Concept

IMPULSE aims to use decommissioned satellites as passive dynamical sensors. The concept relies on the non-maneuverability of the satellites. As decommissioned, these spacecraft do not fire correction or attitude control burns. At LEO (700-1000 km), atmospheric drag is substantially weaker than at lower LEO altitudes, allowing orbital histories to persist over long timescales and providing extended observation records (Picone et al., 2002). Because these spacecraft are no longer intentionally maneuvering, their observed orbital evolution can be modeled primarily through natural perturbations and measurement effects. The measurement that IMPULSE looks for is impulsive events within these histories that cannot be explained after modeling such forces. However, these debris impact events, at a detectable threshold, are rare and carry significant uncertainty when considered individually. Due to this, the pipeline is run on a fleet of satellites so that more observation windows can provide higher statistical power, which would tighten the uncertainty.

2.3 Fleet-Based Experimental Design

The fleet was constructed to maximize statistical power and to target relevant orbital regions. Due to this, the selected orbital altitude band was 700 - 1000 km, to ensure exposure to the dense microdebris population from the Fengyun-1C breakup and the Kosmos-Iridium collision belt (Anz-Meador & Liou, 2010; Liou & Johnson, 2008; Pardini & Anselmo, 2009).. The region's relatively low drag leads to debris persisting for longer periods, which allows the pipeline to use a longer total history for analysis, as well as leading to longer lifespans for the spacecraft themselves. This is also a region of interest due to the heavy use of Sun Synchronous Orbit (SSO) and the low drag (Schild & Noomen, 2022). Therefore, this region provides

relatively high statistical power while being a region of interest, which is favorable for initial IMPULSE testing.

The fleet's individual spacecraft's orbital histories are then taken and segmented into 90-day windows, each of which are run through the detection pipeline. The candidate perturbations from each of these windows across the whole fleet are aggregated as the raw candidate perturbation events.

Although each satellite provides a different orbital history and therefore a different realization of the underlying perturbation environment, the detection pipeline reduces each window to a common measurement. The measurement being whether a calibrated impulsive perturbation trigger above a calibrated threshold occurred. These binary event outcomes are then aggregated across the fleet to estimate a population level trigger rate and its uncertainty. As such, the fleet converts many individually weak measurements into a single statistical measurement of the underlying debris environment.

This data is then fed into the false-positive subtraction procedure to get a tighter count of candidate perturbations.

2.4 IMPULSE Architecture and Decision Framework

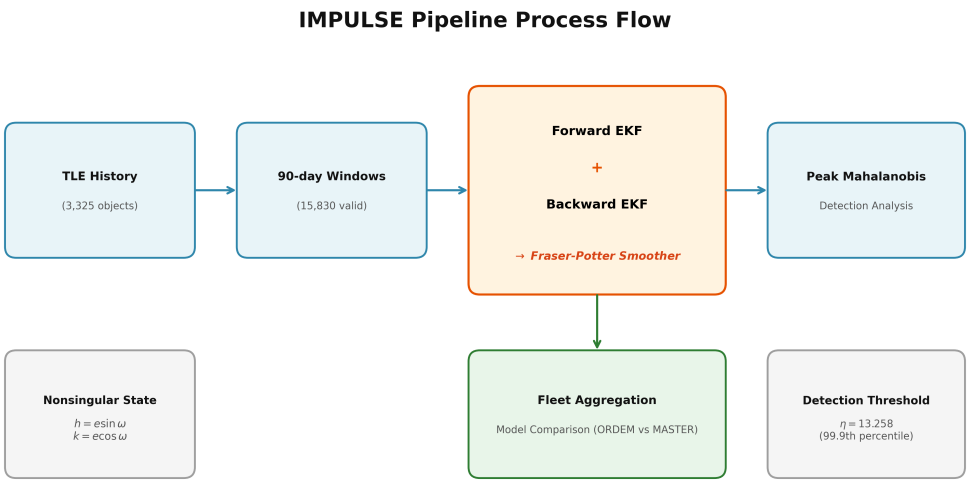


Figure 1: The IMPULSE pipeline. TLE histories for 3,325 objects are segmented into 90-day windows. Each window is processed by a forward EKF, backward EKF, and Fraser-Potter smoother. The peak Mahalanobis distance is tested against a null-calibrated threshold ($\eta = 13.258$, 99.97th percentile). Detections are aggregated across the fleet for model comparison. [AI-generated graphic using Claude (Anthropic), subsequently modified by the student researcher, 2026]

The IMPULSE pipeline starts with the raw TLE histories of decommissioned satellites within 700-1000 km altitude bands sourced from SpaceTrack. These TLE histories are then segmented into 90-day analysis windows. Each of the windows are taken through quality cuts to ensure numerical and statistical stability. This includes the removal of duplicate epochs, anomalous TLEs and sudden discontinuities. The remaining windows are run through a 9-state augmented Extended Kalman Filter (EKF) with non-singular equinoctial elements. The result is smoothed via a fixed-interval Fraser Potter smoother with first-order Gauss Markov

dynamic model compensation.

The history is then studied for its residuals, where candidate impulses are decided by a calibrated threshold. The number of candidate events are aggregated across the fleet to get a final raw candidate perturbation event count. A validation framework provides the model calibrations, thresholds and false-positives which adjust this raw event count to a background-subtracted, smaller candidate event count. This adjusted trigger rate is compared to the expected ORDEM/MASTER event counts to discriminate between them.

The validation and background subtraction steps are therefore a vital piece of the measurement system when considering the highly sensitive environment. They narrow the gap between candidate perturbations and debris induced perturbations. This also means the number of detections, even after adjusted for false-positives and errors, are dependent on the quality of the validation and background-subtraction steps. Thus, detection count alone does not constitute a successful measurement. A large portion of the triggers could be indications of an inadequately characterized background, overly sensitive threshold or systematic error in the dynamical model. The experiment succeeds only if the background-subtracted signal is statistically distinguishable from the quantified false-positive distribution and contains sufficient information to discriminate between ORDEM and MASTER predictions.

3 Orbital-State Estimation

3.1 Orbital-State Representation and Dynamical Model

IMPULSE represents the orbital state in a nonsingular equinoctial nine-element set rather than classical Keplerian elements (Broucke & Cefola, 1972). The six orbital elements are taken as

$$\mathbf{x}_{\text{orb}} = (a, h, k, p, q, \lambda) \tag{1}$$

where $h = e\sin\omega$ and $k = e\cos\omega$, and p, q are the usual equinoctial inclination variables. The remaining three states augment the filter with a first-order Gauss-Markov (FOGM) process that absorbs residual, unmodeled accelerations. This is tuned to primarily mitigate long-term accumulating effects such as atmospheric-density and ballistic coefficient mismatch. The full nine-element state is therefore

$$\mathbf{X} = (\mathbf{x}_{\text{orb}}, W_R, W_S, W_W) \tag{2}$$

The nonsingular substitution via h and k is required by the orbital regime itself, as in classical Gauss variational equations, the rate $d\omega/dt$ contains a factor $1/e$ (Vallado, 2022). As atmospheric drag circularizes the orbits in the SSO regime, e approaches 0, routinely falling below $10^{-4} - 10^{-5}$. At this point, the formulation becomes numerically unstable. Figure 2 shows the coefficient magnitudes versus eccentricity for a representative SSO altitude. The classical $1/e$ term diverges, while the nonsingular rates remain finite and well-conditioned. Since the majority of IMPULSE's fleet lies in low eccentricity orbits due to the drag-dominated nature of the region, this form of representation is necessary for filter stability.

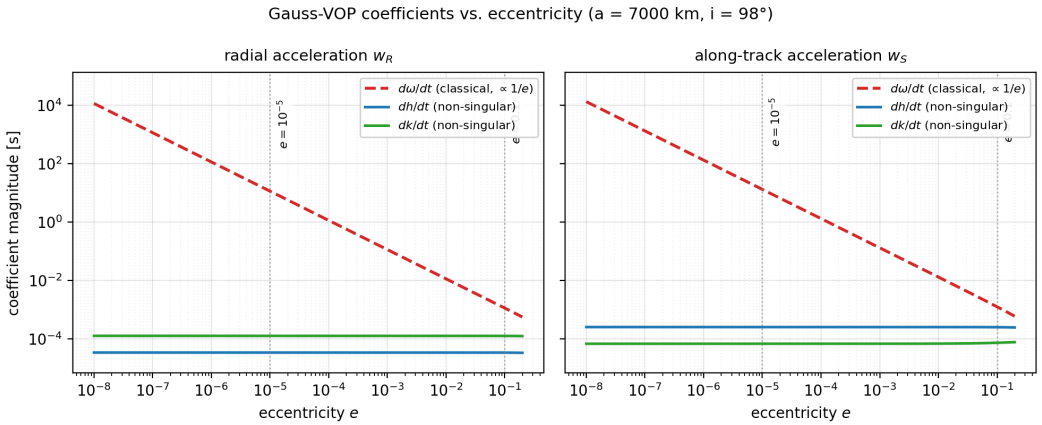


Figure 2: Magnitude of the Gauss-VOP coefficients mapping RSW acceleration into element rates, as a function of eccentricity (reference orbit $a = 7000$ km, $i = 98^\circ$). The classical $d\omega/dt$ rows (dashed red) diverge as $1/e$ as $e \to 0$, while the non-singular (h, k) rows (solid) remain flat and bounded all the way to $e = 0$. [Graphic created by the student researcher using Matplotlib (Python), 2026]

The dynamical model that propagates $\mathbf{x}$ includes the dominant environmental accelerations that can be evaluated through publicly available indices and ephemeris:

- Earth's gravity through J_2
- Atmospheric drag with stochastic C_d, driven by an Ornstein-Uhlenbeck process to model a mean-returning random walk, is used to account for unmodeled variations (Schiemenz et al., 2020; Uhlenbeck & Ornstein, 1930)
- Solar-radiation pressure via a cannon-ball model
- Lunisolar third-body perturbations

All other unmodeled sources of acceleration (such as higher-degree gravity, albedo, thermal radiation, etc.) are absorbed into the FOGM process-noise terms. The measurement then is simply the direct readout of the first six equinoctial elements from each TLE. Measurement noise covariances are taken from published TLE accuracy statistics and are treated as diagonal (Osweiler, 2006). This is a noteworthy limitation of the model, which is partly mitigated by the FOGM and subsequent fixed-interval smoother.

This combination of nonsingular elements, a physically motivated but incomplete force model, and a low-order stochastic acceleration process is the null model against which candidate impulses are later tested.

3.2 Forward and Backward Extended Kalman Filtering

TLE histories only supply discrete, noisy mean-element observations. Recovering a continuous state trajectory, and therefore isolating candidate impulsive events within that trajectory, requires sequential estimation. IMPULSE performs this estimation with two Extended Kalman Filters that operate on the same 90-day window but in opposite directions.

The forward filter is initialized at the first valid TLE of the window and is marched forward in time. At each measurement epoch, the filter predicts the nine-element state under the null dynamical model described in 3.1, computes the innovation, and updates both the state estimate and its covariance. Process-noise intensity is set by the FOGM model parameters tau and q so that slow varying modeling errors are absorbed to avoid false detections.

Once the forward pass reaches the end, a second filter is initialized at the terminal state and covariance is run backward. The backward filter is able to incorporate future states in its predictions that were unavailable to the forward pass. Because the two filters process the data in opposite chronological order, their estimates at any interior epoch are independent given the measurement. Any large difference between the two cannot be explained by ordinary process or measurement noise under the null model.

The forward filter can detect an abrupt change only after it has occurred and only relative to the preceding data. The backward filter sees the same change relative to the subsequent data. Their mutual inconsistency, which is normalized by joint uncertainty among the two, is therefore a sensitive indicator of a local dynamical discontinuity. This discrepancy becomes the detection statistic.

3.3 Fraser-Potter Smoothing and Process Noise

As established, the detection statistic must isolate a localized dynamical inconsistency. Due to this, both past and future information are required at each epoch. IMPULSE therefore fuses the two EKF passes with a Fraser-Potter smoother (Fraser & Potter, 1969). At each epoch, the smoothed state is:

$$\mathbf{x}_S = W\mathbf{x}_F + (I - W)\mathbf{x}_B, \qquad W = P_B(P_F + P_B)^{-1} \tag{3}$$

where subscripts F and B denote the forward a posteriori and backward a priori quantities respectively. The associated smoothed covariance is formed in the symmetric Joseph-style expression. The smoother gives the minimum-variance estimate of the entire nine-element state history given all the measurements inside the 90-day window.

The three unmodeled-acceleration states are driven by a first-order Gauss-Markov (FOGM) process:

$$\dot{\mathbf{w}} = -\frac{\mathbf{w}}{\tau} + \boldsymbol{\nu}, \qquad \langle \boldsymbol{\nu}(t)\boldsymbol{\nu}(t')^\top \rangle = q\,\delta(t - t')\,\mathbf{I} \tag{4}$$

with fixed parameters $\tau = 3 \times 10^4\,\mathrm{s}$ and $q = 10^{-19}$. These values are set by the validation ladder so that slow model mismatch (density errors, ballistic-coefficient drift) is absorbed by the FOGM prior rather than interpreted as impulsive events (Schiemenz et al., 2020; Uhlenbeck & Ornstein, 1930). The resulting detection statistic is the peak Mahalanobis distance between the forward and backward estimates, evaluated with the smoother covariance (Bennett et al., 2023).

3.4 Candidate Impulse Detection

The detection statistic used is the peak Mahalanobis distance between the forward and backward state estimates, evaluated with the smoother covariance:

$$d_M = \sqrt{(\mathbf{x}_F - \mathbf{x}_B)^\top P_S^{-1} (\mathbf{x}_F - \mathbf{x}_B)}. \tag{5}$$

A window is declared triggered when the peak d_M exceeds the pre-calibrated threshold $\eta = 13.258$ (99.97th percentile of the synthetic null distribution). An optional secondary metric for diagnostic and sanity-check is the along-track acceleration component $|W_s|$. Large, localized spikes in W_s frequently point to a true impulse, as debris impacts are largely along-track, but the primary decision remains within the Mahalanobis distance. It is important to clarify three intertwined claims:

- A **statistical trigger** is simply a window whose $d_M > \eta$.
- A **recovered synthetic impulse** is a trigger that coincides with a known injected Δv (used in calibration tests only).
- A **confirmed debris impact** would require additional post-processing, background subtraction, or other additional attribution that isolates the event from non-debris systematic noise.

V1 reports only the first of the three. The background-subtraction and model-comparison stages (Sections 6-7) quantify how much of the statistical trigger population can be attributed to debris versus residual TLE systematics to inch closer to confirmed debris impacts.

4 Validation Framework

4.1 Validation Strategy

A statistical trigger is a candidate orbital perturbation. It cannot automatically be interpreted as debris. Multiple non-debris mechanisms can generate statistically significant residuals. These include estimator transients, incompleteness in the force model, unmodeled physical effects (density spikes, residual SRP), and TLE-specific artifacts. IMPULSE therefore runs through a staged validation ladder that progresses from fully controlled synthetic conditions to real TLE data.

Table 1: 4-stage Validation Ladder

Gate	Question	Purpose
1. Null test	Does the filter trigger under ideal conditions?	Bound intrinsic false-positive rate
2. Bias recovery	Can it track slowly varying forces?	Verify FOGM absorbs model error
3. Strike injection	Can it recover known impulses?	Measure detection efficiency
3b. Real-TLE injection	Does sensitivity persist on real TLEs?	Test under observational noise
4. Clean vs. debris	Does it distinguish a realistic debris population?	Confirm population-level sensitivity

Table created by the student author using LaTeX, 2026

Only after these controlled tests does the pipeline move to the scrambled-TLE null that quantifies the final empirical false-positive floor under realistic catalog noise.

4.2 Synthetic, Null, and Real-TLE Validation

Synthetic impulse injection places a known Δv of controlled magnitude and epoch into an otherwise clean trajectory and runs the complete detection pipeline. Recovery is declared only if the peak Mahalanobis distance exceeds $\eta = 13.258$ inside a ± 1-day window of the true epoch.

Complementary null tests operate without injected impulse. The intrinsic null (clean synthetic data) measures the rate at which the estimator itself triggers a detection. The confounder null deliberately injects density variations, C_d drift, and geomagnetic activity to quantify the false-positive rate at which these effects can mimic a debris impulse. Low rates on both nulls confirm that the pipeline does not independently invent detection under the controlled conditions.

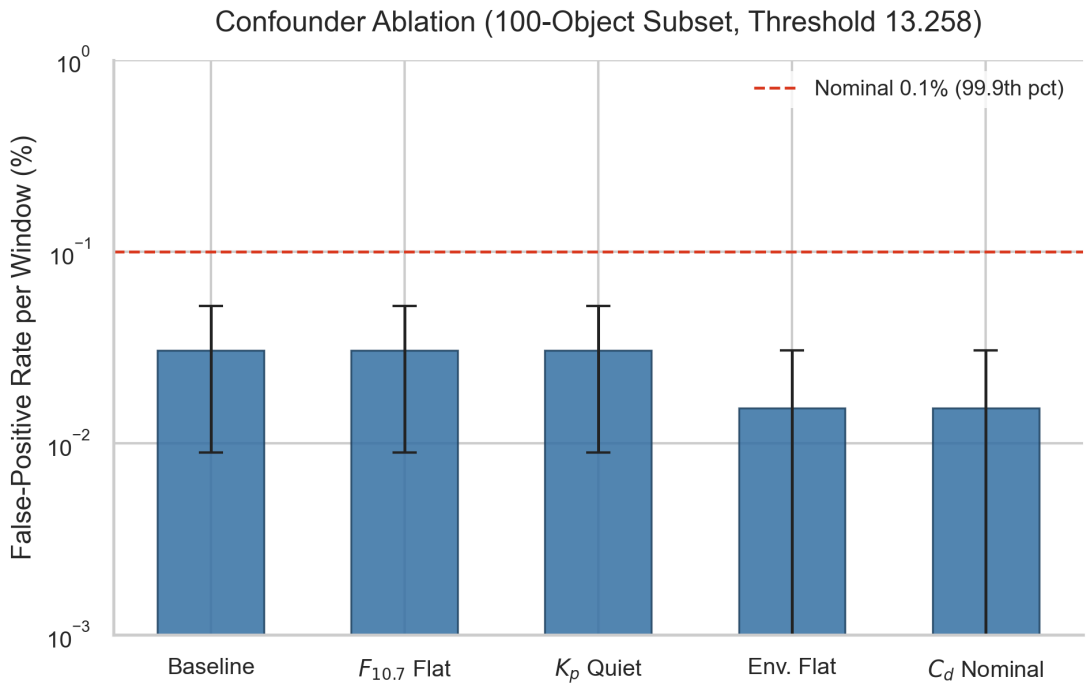


Figure 3: False-positive rate across individual filter-degradation variants on the 100-object subset ($N \approx 6{,}530$ windows per variant). Error bars indicate 95% binomial confidence intervals. All synthetic physical confounders remain strictly below the nominal 0.1% threshold (dashed line) and show no statistically significant deviation from the baseline rate. [Graphic created by the student researcher using Matplotlib (Python), 2026]

Synthetic validation alone is useful for quantifying the model's error, but it cannot establish reliability on real catalog data. TLE residuals contain temporally coherent structures (station changes, unmodeled maneuvers, density mismodeling) which are absent from the synthetic null model. The scrambled-residual null is designed to address this gap partially. A smooth spline is fit to each orbital-element time series, the residuals are permuted randomly, and the detection pipeline is re-run. Scrambling residuals preserve the residual amplitude and overall noise statistics while destroying temporal coherence and any genuine

dynamical signatures. The resulting trigger rate (19.46%) therefore quantifies the false-positive contribution arising from temporally coherent TLE-associated systematics alone.

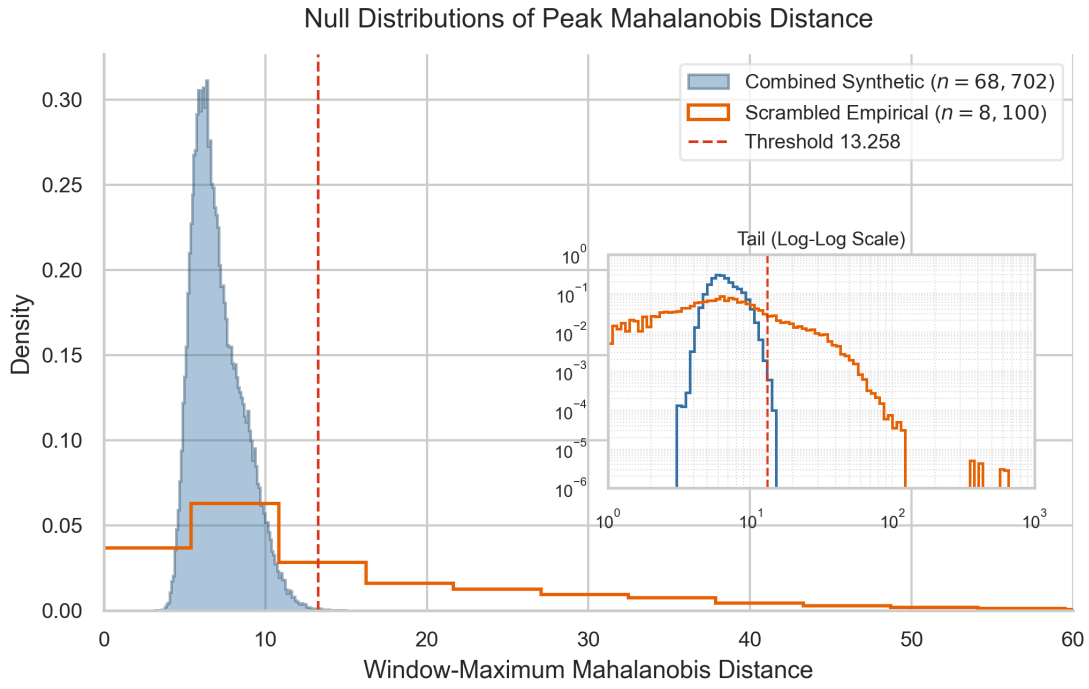


Figure 4: Null distribution of peak Mahalanobis distances from the 68,702-window combined confounder run (histogram with step-filled density). Vertical dashed line indicates the 99.97th percentile threshold at 13.258. Right-side inset shows the heavy-tail distribution on a log-log scale with the scrambled empirical null overlaid for comparison. [Graphic created by the student researcher using Matplotlib (Python), 2026]

This empirical background must be subtracted from the real fleet before any debris signal can be claimed.

4.3 Estimator Characterization and Decision Criterion

The bias-recovery tests confirm that a constant acceleration of 10^{-6} m/s^2 is tracked without generating false triggers. The FOGM parameters τ and q that resulted in this were $\tau = 3 \times 10^4$ and $q = 10^{-19}$. These parameters were tuned via a binary grid-search algorithm that converged at the aforementioned values. The parameters were then also tested for impulsive event sensitivity, explored in Section 5, to ensure they were unabsorbed by the FOGM model.

The pre-specified model-discrimination criterion requires $|\log_{10} B| > 1$ to declare decisive statistical evidence between competing hypotheses (Kass & Raftery, 1995). Detection efficiency $\epsilon(\Delta v)$ and measured false-positive rates enter the calculation of expected event counts under ORDEM and MASTER. The observed net excess is then compared against those predictions. Discrimination is declared only if the residual after the background subtraction exceeds the quantified systematic floor required by the Bayes factor.

5 Impulse Detectability and Recovery

5.1 Synthetic Impulse Recovery

Controlled Δv impulses were injected along-track into synthetic clean trajectories across a logarithmic grid from 10^{-5} to 10^{-1} m/s. The full detection pipeline was then run on the impulse-injected data, and recovery was declared only if the peak Mahalanobis distance exceeded the threshold $\eta = 13.258$ within a $\pm$1-day window.

The resulting efficiency curve rises sharply between 10^{-3} and 10^{-1} m/s (Fig. 6). It flattens out after about the 0.1 m/s mark. The 50% detection threshold on the synthetic data is

$$\Delta v_{50} = 0.041\,\mathrm{m\,s^{-1}}. \tag{6}$$

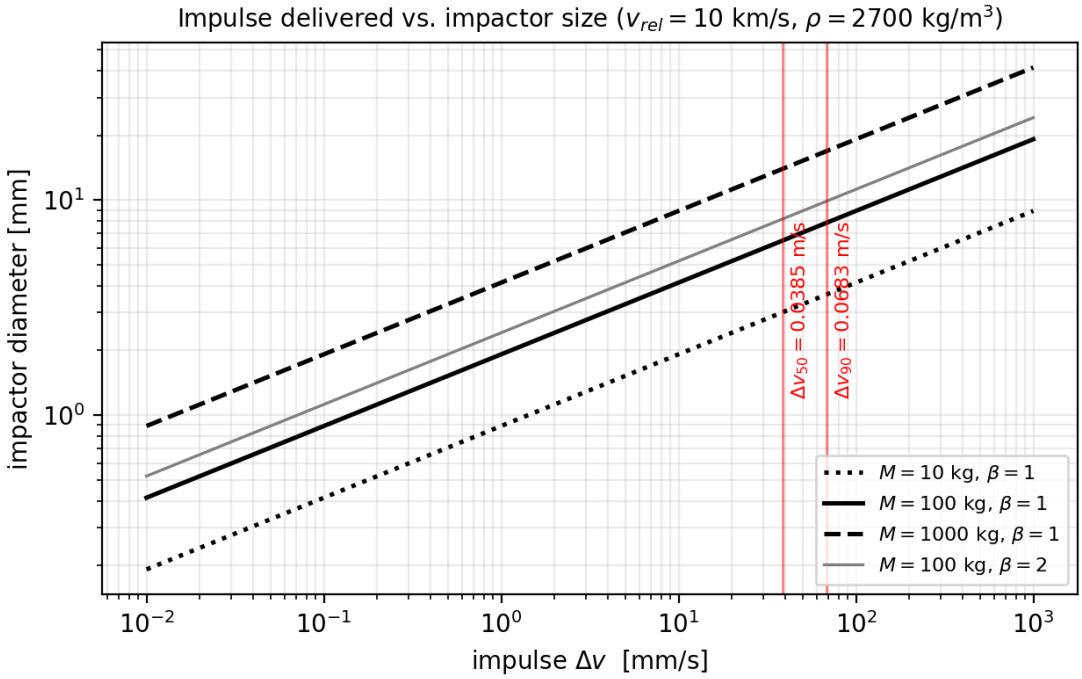


Figure 5: Impulse delivered by a hypervelocity impact as a function of impactor diameter, for spacecraft masses of 10, 100, and 1000 kg at a 10 km/s relative velocity (perfectly inelastic, $\beta = 1$, solid black curves; $\beta = 2$ momentum enhancement shown in gray for the 100-kg case) (Christiansen, 2009). Vertical lines mark the fitted Gate 3b sensitivity Δv_{50} and Δv_{90}. [Graphic created by the student researcher using Matplotlib (Python), 2026]

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This value determines the sensitivity of the measurement pipeline. It is important to note that this is under ideal, synthetically clean, conditions. At $0.041\,\mathrm{m\,s^{-1}}$, the pipeline is expected to detect the debris impulse 50% of the time. This efficiency rate later determines the number of expected debris-induced candidate perturbation statistical trigger events for ORDEM and MASTER scenarios. It is important to note that the 50% rate is not a false-positive; it is the proportion of expected missed debris events during

the detection pipeline, while false-positive rates quantify the proportion of statistical triggers that are not induced by debris impacts.

5.2 Synthetic and Real-TLE Performance

The same synthetically injected impulse grid was injected into real TLE windows with scrambled residual to test the sensitivity of the pipeline in the presence of TLE-induced observational noise and catalog artifacts. The real-TLE efficiency curve yields a Δv_{50} at a slightly lower threshold than the synthetic:

$$\Delta v_{50} = 0.0385\,\mathrm{m\,s^{-1}}, \tag{7}$$

This is consistent, as the real TLE data is pre-processed with quality cuts that remove problematic windows.

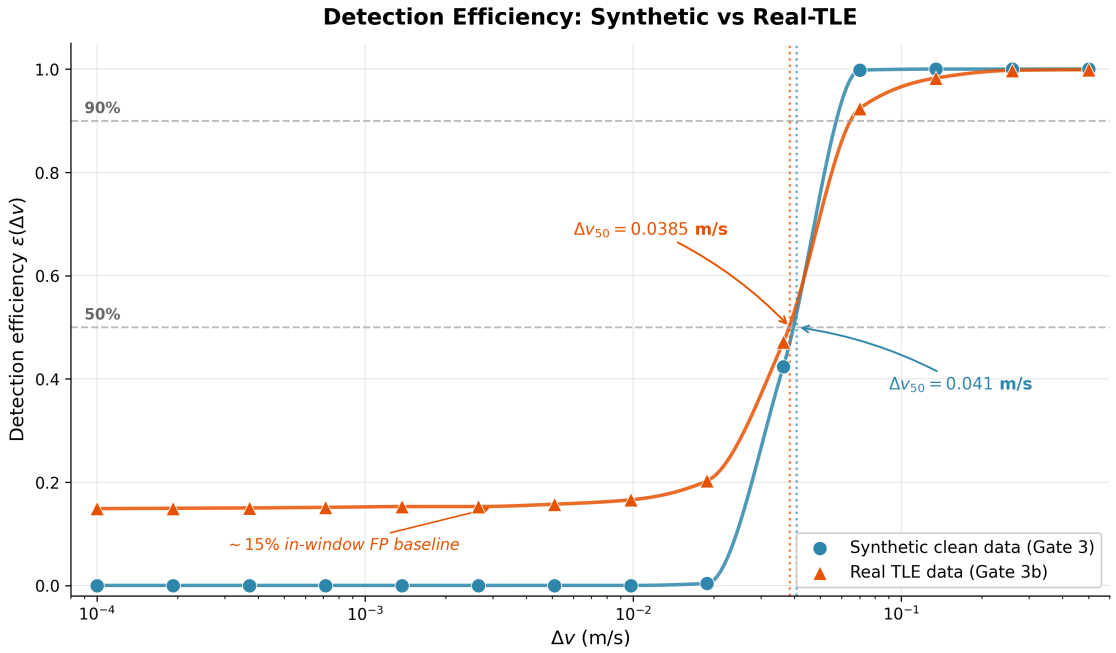


Figure 6: Detection efficiency $\epsilon(\Delta v)$ for synthetic clean data (blue circles, Gate 3) and real-TLE injection (orange triangles, Gate 3b). The real-TLE $\Delta v_{50} = 0.0385\,\mathrm{m/s}$ is slightly better than the synthetic $\Delta v_{50} = 0.041\,\mathrm{m/s}$ consistent with quality-cut pre-filtering. The $\sim$15% baseline at low Δv on real TLEs is the in-window false-positive rate from noise. [Graphic created by the student researcher using Matplotlib (Python), 2026]

However, this threshold cannot be trusted at face value. At negligible impulse magnitudes, the real-TLE efficiency curve shows a detection efficiency of $\sim 15\%$. This baseline is the probability that a real window already contains an artificial peak inside the recovery interval and is therefore the in-window false-positive rate under observational noise. It is important to note that this is a subset of the scrambled-residual real-TLE null quantification of the false-positive rate of 19.46%. The contrast between the near-zero synthetic floor and 15% scrambled residual real-TLE floor is the first quantitative indication that the TLE cataloging systematics dominate the measurement.

5.3 Expected Observable Signals Under ORDEM and MASTER

The raw debris flux predictions from NASA's ORDEM and MASTER cannot directly be compared with the observed trigger population. Each model's fragment distribution must be propagated by IMPULSE's selection function. This includes passing through the detection-efficiency curve $\epsilon(\Delta v)$, per-object observation duration, and the specified geometric cross-section. Through Monte Carlo sampling, the two models yield the following detectable events under IMPULSE (Braun, Horstmann, et al., 2021; Krisko, 2014):

Table 2: Hypothesized Predicted Detections by Model

Model	Expected impacts	Predicted detections	Effective ε
ORDEM	131,997	847.9	0.64%
MASTER	968.9	6.1	0.63%

Table created by the student author using LaTeX, 2026

As noted earlier, the two models disagree by $\sim 138\times$, with NASA's ORDEM predicting significantly higher expected impacts than ESA's MASTER. Even after considering different efficiency curves for each model, the ratio remains. The detection efficiencies are largely similar because the factor that contributes most to it, the debris fragmentation model, is the same NSBM (NASA Standard Breakup Model) model across the two (Johnson et al., 2001; Reynolds et al., 1998). The significantly low efficiency and predicted detections also highlight IMPULSE's limits: it is not suitable for standalone event debris detection, but rather it is a discrimination pipeline. The 847.9 versus 6.1 detectable event discrepancy sets the scale against which the later fleet-scale excess must be judged.

6 Systematic False Positive Characterization

6.1 Intrinsic and Confounder False Positives

The IMPULSE model was validated in a controlled synthetic condition to ensure the lack of hallucinated false-positives. The intrinsic null test ran the detection pipeline on synthetically clean orbital history data and quantified the model's intrinsic false-positive rates under ideal conditions. It yielded a trigger rate of only 0.06% at the adopted threshold $\eta = 13.258$. When realistic but non-impact perturbations were deliberately injected into a synthetically clean null, the false-positive rate remained below 0.03% (Figure 3). The confounder quantifies the uncertainty for solar radiation pressure, geomagnetic K_p, and the nominal C_d separately, as well as all three together. It does so by holding two of them at a degraded baseline: $F_{10.7} = 150.0$, $K_p = 2.0$, $C_d = 2.2$. By fixing the values for two, and letting the other influence the detection pipeline, the resulting discrepancy can be attributed solely to the varying force. To quantify the total false-positive rate via interaction effects between the three, a run with all three fixed versus a run with all three varying is used. Through this confounder test, the following false-positive rates were quantified:

The relatively low false-positive rates suggest that the force model is effectively being absorbed by the FOGM model and that it is sufficiently and quantifiably effective for the purposes of IMPULSE. This implies

Table 3: Confounder Test Results Summary

Filter assumption	Windows	Triggers	FP rate (%)	Mean peak D_M
Matched baseline	6,526	2	0.031	7.11
Flat $F_{10.7}$	6,528	2	0.031	7.13
Quiet K_p	6,532	2	0.031	7.15
Flat environment (all)	6,529	1	0.015	7.16
Nominal C_d	6,531	1	0.015	7.11

Table created by the student author using LaTeX, 2026

that any major noise floor or errors in the model likely does not result from the force model or FOGM model, but rather from the real-TLE data quality and other systematic errors.

6.2 Scrambled-TLE False Positives

The synthetic and confounding tests establish that the pipeline's force, propagation, and FOGM model are effective in not hallucinating detections when the dynamical model is correct. However, real-TLE histories produce a dramatically higher trigger rate when run on the full fleet. When the detection pipeline is applied to the full set of 15,830 valid windows, 8,161 peaks exceed $\eta = 13.258$, corresponding to a raw trigger rate of 51.55%.

The exceedingly high rate does not fit within the predicted value for ORDEM (847.9 triggers) or MASTER (6.1 triggers). The false-positive rates quantified by the synthetic clean null (0.06%) and the confounder null ($< 0.03\%$) tests cannot explain this rate. The discrepancy indicates the presence of real-TLE residuals that are absent from the synthetic truth model.

To isolate the contribution of these structures partially, a scrambled-residual null was constructed. A smooth cubic spline is first fit to each orbital-element time series. The residuals about the spline are then randomly permuted. This constitutes the new orbital history data that the complete IMPULSE pipeline is run through. The scrambled data preserves the amplitude distribution and overall noise statistics of the residuals, while destroying temporal coherence and any genuine debris impacts such that they are not counted in the false-positive quantification.

Table 4: Scrambled FP Trigger Rate

Metric	Value
Valid windows	15,830
Triggers ($> \eta$)	3,081
Scrambled FP rate	19.46%

Table created by the student author using LaTeX, 2026

The scrambled test yields 3,081 triggers out of 15,830 windows, or a false-positive rate of 19.46% (Table 4). The gap between the synthetic floor (0.06%) and the scrambled-TLE floor (19.46%) is itself a primary result. Over two orders of magnitude separate the idealized false-positive rate from the rate

measured when real and imperfect catalog data was introduced. As such, approximately 38% of the raw detections $(3,081/8,161)$ can be attributed to temporally non-coherent TLE systematics alone. This noise floor must be investigated for residual systematic contributions before any debris interpretation is possible.

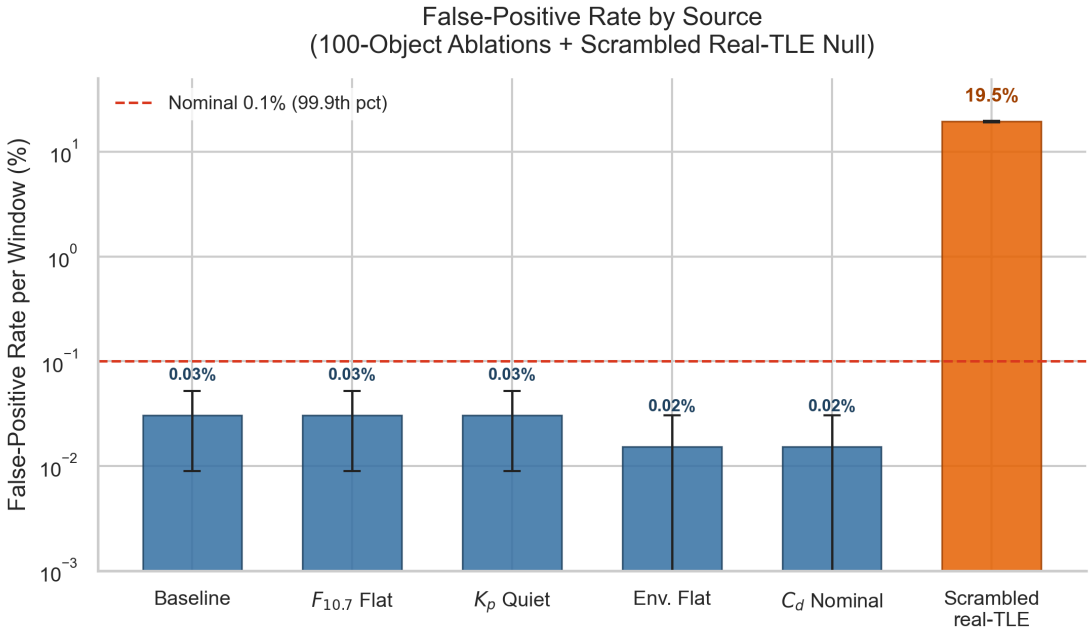


Figure 7: False-positive rate across individual filter-degradation variants on the 100-object subset ($N \approx 6{,}530$ windows per variant). Error bars indicate 95% binomial confidence intervals. All synthetic physical confounders remain strictly below the nominal 0.1% threshold (dashed line) and show no statistically significant deviation from the baseline rate. In contrast, the scrambled-residual real-TLE test quantifies a significantly higher false-positive budget. [Graphic created by the student researcher using Matplotlib (Python), 2026]

7 Fleet-Scale Analysis

7.1 Fleet Construction and Observation Windows

The initial fleet consisted of all decommissioned spacecraft with Space-Track TLE coverage between 2009 and 2026 whose mean altitude lay inside the 700-1000 km band. The cut-off point for TLE history was set at 2009 as it denoted the post-Fengyun-1C breakup event, which dramatically changes the flux in the region. Objects that were still operational or exhibited station-keeping behavior were deliberately removed from the fleet. The final catalog contained 3,325 decommissioned spacecraft.

Each object's TLE history was segmented into successive, non-overlapping 90-day windows. Four quality cuts were performed:

- Minimum of 15 TLEs inside the window.
- Any data discontinuities were trimmed out.
- Altitude changes inside the window must be less than 10 km between TLEs, as it indicates station-changing or other cataloging artifacts.

- Exclusion of the first window for objects whose total span exceeds 90 days in order to account for filter transients.

Of the original 22,888 raw windows, 15,830 (69.2%) survived the cuts. For each window, a 7-day edge trim was imposed to suppress filter transients and smoother boundary effects. This left around 76 days of usable detection time per window. The resulting dataset therefore consists of 15,830 independent statistical trials.

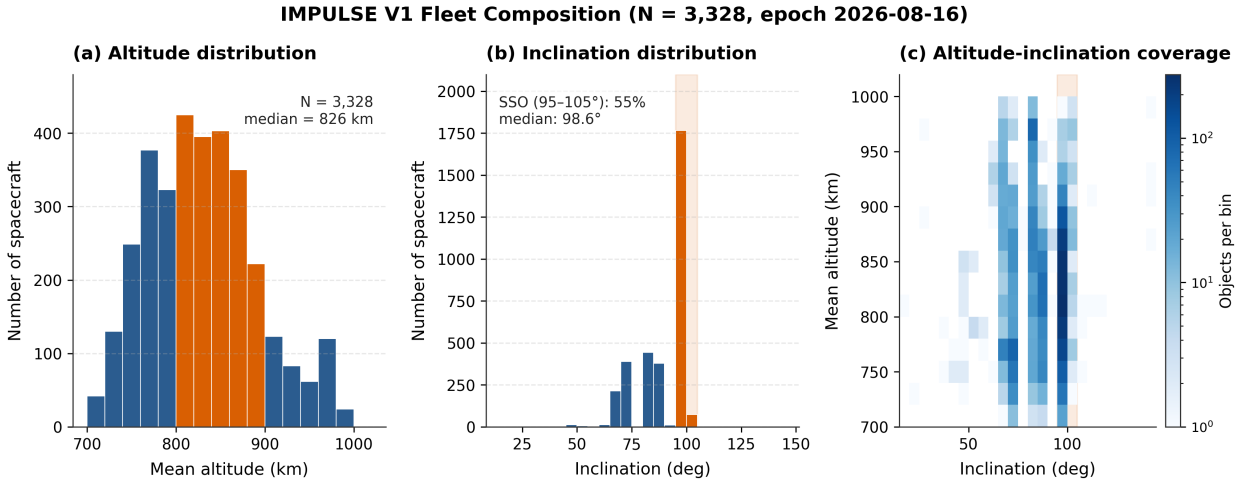


Figure 8: IMPULSE V1 fleet composition at the final catalog epoch (2026-08-16). (a) Mean-altitude distribution; all objects lie within the 700-1000 km selection band. (b) Inclination distribution; the sun-synchronous shell (95-105°, orange) contains 55% of the fleet. (c) Joint altitude-inclination coverage (log color scale), showing the orbital regimes sampled by the selection function. [Graphic created by the student researcher using Matplotlib (Python), 2026]

The altitude distribution peaks around 820 km and 880 km. A total of 55% of the objects occupy the sun-synchronous inclination shell (95°-105°). Observation spans range from a single valid window to 17 years of continuous coverage. Figure 8 summarizes the three-panel fleet composition regarding altitude, inclination, and joint altitude-inclination coverage. The strong concentration in the SSO regime is consistent, as it is a widely used commercial regime.

7.2 Fleet Characteristics and Raw Detection Statistics

The complete pipeline was executed on this set via memory-safe multiprocessing. Summary statistics are detailed in Table 5.

When the calibrated pipeline is applied to the 15,830 valid windows, 8,161 peaks exceed the threshold $\eta = 13.258$, corresponding to a raw trigger rate of 51.55%. Over half of all windows therefore contain temporally coherent signatures that exceed the detection threshold $\eta = 13.258$. The peak Mahalanobis distance across all windows is 19.44, and the distribution is strongly right-skewed. This excess is the starting point for the background subtraction and model comparison that follow using the false-positive contribution quantified in Section 6.

Table 5: Full Fleet Run Summary Statistics

Metric	Value
Total objects processed	3,325
Valid analysis windows	15,830
Peaks computed	15,786
Mean peak Mahalanobis distance	19.44
Threshold η (99.97^{th} percentile)	13.258
Raw detections ($> \eta$)	8,161
Raw trigger rate	51.55%

Table created by the student author using LaTeX, 2026

7.3 Background Subtraction and the Systematic Floor

The background subtraction can be performed by direct difference:

$$N_{\text{net}} = N_{\text{obs}} - N_{\text{FP}} = 8,161 - 3,081 = 5,080 \pm 103 \tag{8}$$

This is about $32.09\% \pm 0.6\%$ of all windows. This net excess is the population of temporally coherent real TLE data anomalies that survive the scrambled-TLE null. It is not yet a debris count, but rather the sum of all residual systematics as well as the genuine debris impact signatures that the null model cannot remove.

The $\sim 15\%$ in-window false-positive rate quantified earlier during real-TLE impulse injection is consistent with, but not identical to, the 19.46% scrambled rate. The injection floor measures the probability that a real window already contains an artificial peak inside a narrow recovery interval. The scrambled rate measures the broader population of coherent structures across the entire window. Together, they indicate that a substantial fraction of the candidate population is a result of TLE-related systematics rather than external impulsive events.

This net excess can be decomposed under each debris environment hypothesis. Adding the model-predicted triggers to the scrambled false-positive floor produces:

Table 6: Residual Excess of ORDEM and MASTER

Scenario	Predicted total	Observed	Residual excess
ORDEM + scrambled FP	3,929	8,161	$+4,232$
MASTER + scrambled FP	3,087	8,161	$+5,074$

Table created by the student author using LaTeX, 2026

Expressed as percentages, the decomposition comprises 19.5% scrambled TLE noise (measured) and a 5.4% ORDEM debris contribution, leaving an implied non-debris systematic floor of 26.7% (or 32.0% under the MASTER hypothesis).

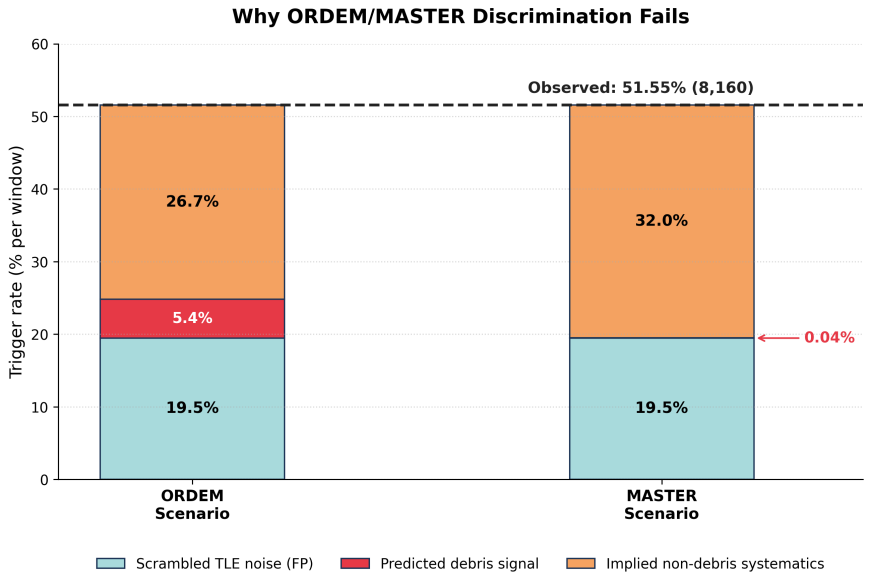


Figure 9: Decomposition of the observed 51.55% trigger rate. The scrambled TLE noise contributes 19.5% (measured). Under the ORDEM scenario, debris contributes 5.4% and the implied systematic floor is 26.7%. Under MASTER, debris is 0.04% and the implied systematic is 32.0%. In both cases the systematic floor dwarfs the debris signal, blocking model discrimination. [Graphic created by the student researcher using Matplotlib (Python), 2026]

7.4 Comparison With ORDEM and MASTER

When the net excess of 5,080 events (32.09%) is compared with the selection-function-propagated predictions from Section 5.3 (847.9 detectable events under ORDEM and 6.1 under MASTER), both models under-predict the observation. After adding each prediction to the scrambled false-positive floor, the residual excess is several times larger than either debris signal.

The 26-32% systematic floor is the central quantitative result of IMPULSE V1. It establishes a clear information limit for TLE-based passive sensing under the baseline operating regime. The detection pipeline has been tested for sensitivity and calibration, suggesting the limitation is not algorithmic. The limitation is the temporally coherent non-debris systematic structures present in real TLE histories.

In both ORDEM's (5.4%) and MASTER's (0.04%) hypothesized debris contributions, the debris difference of 5.3 percentage points is smaller than the uncertainty attached to the systematic component. The Bayes factor therefore falls inside the indifference zone $[-1, +1]$ and no model discrimination is possible.

This is not evidence that either ORDEM or MASTER is incorrect. It is evidence that the present IMPULSE measurement system does not properly account for the temporally coherent non-debris real-TLE systematic structures and therefore cannot yet isolate the debris contribution. The observed excess must be interpreted strictly as an upper bound on the combined debris-plus-systematic rate rather than as a measurement of the debris population itself.

8 Discussion

8.1 What IMPULSE Demonstrated

IMPULSE V1 demonstrates that passive dynamical sensing using TLE histories is a technically viable concept for detecting anomalous structures in orbital histories. A 9-state equinoctial EKF-smoother pipeline can be made numerically stable on real TLE data and recovers impulses at $\Delta v_{50} = 0.041\,\mathrm{m\,s^{-1}}$ (synthetic) and $0.0385\,\mathrm{m\,s^{-1}}$ (real TLEs) with 50% detection efficiency, producing a well-calibrated detection statistic. The four-stage validation ladder confirms that ordinary force-model error and estimator instability contribute negligibly to the false-positive budget, indicating that the present force model is adequate and that the FOGM process successfully absorbs slow-decaying mismodeling such as drag and solar-radiation-pressure residuals.

When the same pipeline is applied to 3,325 decommissioned spacecraft ($15,830$ valid windows), it returns a large population of temporally coherent orbital anomalies (raw trigger rate 51.55%). After subtraction of the scrambled-TLE background, a net excess of $5,080 \pm 103$ events (32.09%) remains.

IMPULSE also provides the first large-scale test of the dual-EKF + Fraser-Potter + FOGM architecture explored by Bennett et al. (2023) on real TLE mean-element histories, demonstrating numerical stability and sensitivity while revealing residual catalog systematics as the dominant limitation.

8.2 Why Model Discrimination Failed and the Information limit

The model discrimination goal was not achieved in V1 because of an information limit that is not currently decomposable under the IMPULSE V1 operating baseline model. The expected debris signals after selection-function convolution are only 5.36% (ORDEM) and 0.04% (MASTER). Both are substantially smaller than the 26-32% non-debris systematic floor that remains after the strongest empirical null test currently available. The debris difference between the two models (5.3 percentage points) is therefore smaller than the uncertainty attached to the background floor, and the pre-specified Bayes-factor criterion is not satisfied.

This negative result quantifies the practical information limit of TLE-based passive sensing under the baseline operating regime. It also establishes the additional noise introduced when comparing the pipeline on injected synthetic impulses versus real TLE data. The null tests establish that the pipeline is sensitive; the data simply do not contain debris signatures that can be isolated from residual temporally coherent catalog systematics at the required confidence level. Establishing that boundary is necessary for charting a path forward before any stronger claims can be made.

8.3 Limitations and Path Forward

The dominant limitations are extrinsic to the estimator. It is the temporally coherent structure present in real TLE data that survives the scrambled-residual real-TLE null test. Intrinsic limitations (force-model incompleteness, FOGM tuning, measurement-noise assumptions) have been shown via null tests to be secondary.

Therefore, in order to achieve model discrimination, it is important to reduce the systematic floor via

decomposing the sources of anomalies in the real TLE data. One promising option is to test object-type stratification, testing rocket-bodies versus payloads. As rocket bodies cannot maneuver, this could characterize the false-positives contributed by falsely labeled decommissioned spacecraft or spacecraft that continue to do small correction burns after decommissioning (Li & Chen, 2018). Correlating the spikes in the Mahalanobis distance with geomagnetic activities could help isolate events that were too strong for the FOGM model to absorb, and were instead counted as statistical triggers. Additionally, temporal monitoring across the solar cycle can be a useful gate to investigate if the Mahalanobis distance peaks in accordance to any temporally recurring events or phenomena. These steps deliberately do not involve making modifications to the measurement pipeline, as the pipeline is shown to be sensitive to impulsive events. They require a more refined decomposition of the existing V1 dataset. That said, it is important to test model-intrinsic modifications, such as higher order filters or implementing a more sophisticated atmospheric model, as they may introduce unexpected errors when working with real-TLE data that was not present in synthetic null tests. If the systematic floor can be lowered by approximately a factor of three, the ORDEM signal would rise above the residual background and discrimination would become possible. That is the concrete objective of V2.

9 Conclusion

IMPULSE (Inferred Microdebris Population from Unmaneuvered LEO State Evolution) set out to test whether historical orbital histories of decommissioned LEO spacecraft contain enough information to discriminate between the microdebris predictions of NASA's ORDEM 3.0 and ESA's MASTER 8, which disagree by $138\times$ in the orbital debris flux between 700-1000 km altitudes. A 9-state equinoctial EKF-Smoother pipeline with first-order Gauss-Markov dynamic model compensation was constructed, calibrated, and validated through a four-stage testing ladder of null tests, bias recovery, impulse injection, and clean-versus-debris experiments. On both synthetic and real TLE data, the pipeline recovers impulses near $\Delta v_{50} = 0.04\,\mathrm{m\,s^{-1}}$ and produces a well-characterized debris statistic.

Applied to a fleet of $3,325$ decommissioned spacecraft ($15,830$ valid analysis windows), the measurement system yields a raw trigger rate of 51.55%. After background subtraction of the false-positive budget quantified by the scrambled-residual real-TLE null test (19.46%), a net excess of $5,080 \pm 103$ events (32.09%) remains. This net excess is dominated by a non-debris systematic floor of 26%-32%, which is several times larger than the expected ORDEM debris signals (5.36%) and orders of magnitude greater than the MASTER predicted signal (0.04%). Consequently, the prespecified Bayes-factor criterion is not met and model discrimination is not achieved.

This negative result quantifies the information limit of TLE-based passive sensing: although the detection architecture is sensitive to impulsive events, residual temporally coherent catalog systematics overwhelm the microdebris contribution. Reducing the systematic floor through object stratification, geomagnetic correlation, temporal correlation, and improved filtering is the necessary next step. The V2 plan shifts its attention from constructing the detection pipeline to decomposing this systematic noise floor in order to isolate the debris-induced impulses and lead to discrimination of competing debris-environment models.

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